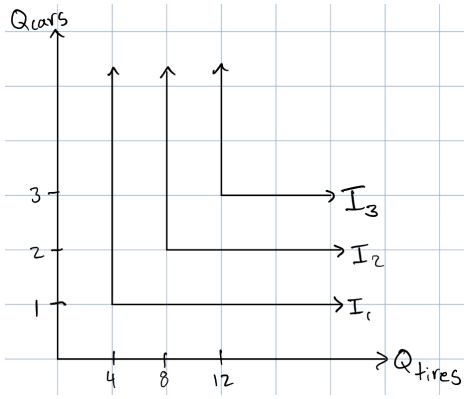
# Ch. 22 CW #2 Solutions

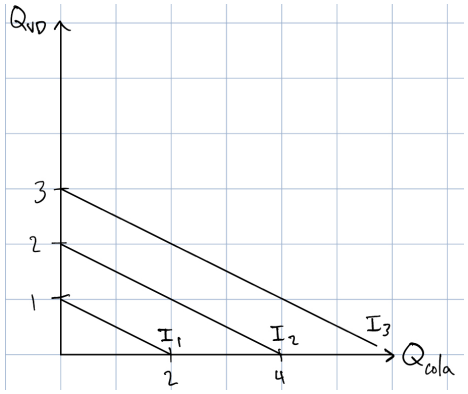
Principles of Microeconomics

November 18, 2025

1. (a) Since cars and tires are perfect complements, Isabella's indifference curves create right angles.

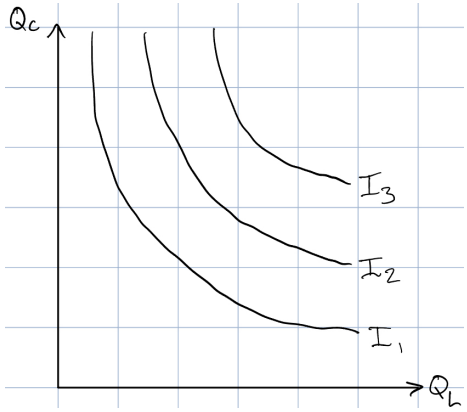



(b) Since Isabella gets utility only from her caffeine intake, she's always willing to trade off Valley Dew and cola at the same ratio, 1:2. That means they're perfect substitutes and her indifference curves are straight.

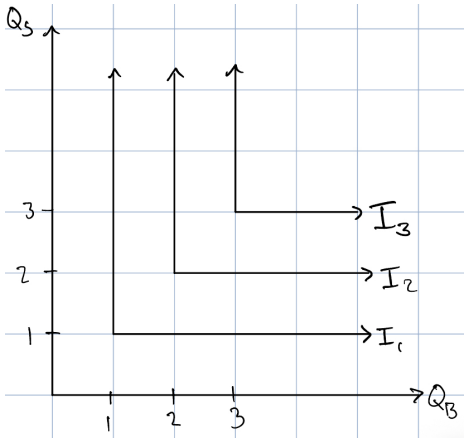



(c) Since both have diminishing marginal utility, her MRS will decrease as you move from left to right on her indifference curves. That means

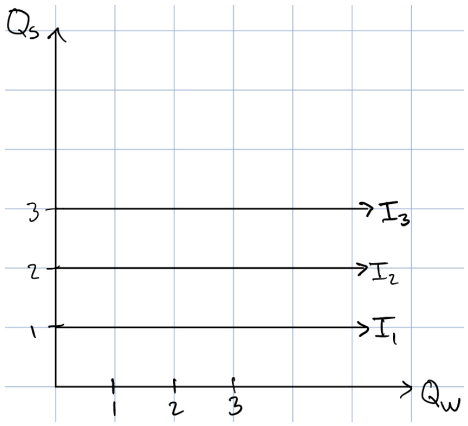
her indifference curves are curved.




(d) Since she always wants to buy skis and bindings together, they are perfect complements, so her indifference curves create right angles.




(e) Since water doesn't affect her utility, her indifference curves are horizontal lines.




2. Diminishing marginal utility means that Marta gets less additional utility from each additional cup. That means her first cup gives her the most additional utility and her last cup gives her the least.

3. (a) The bundle gives Bridget and Erin utility levels of

$$U_B(10,5) = 10(10)(5) = 500$$
$$U_E(10,5) = 0.2(10^2)(5^2) = 500.$$

The bundles that give Bridget the same utility are defined by

$$500 = 10FC$$
$$50 = FC$$
$$C = \frac{50}{F}.$$

The bundles that give Erin the same utility are defined by

$$500 = 0.2F^2C^2$$
$$2,500 = F^2C^2$$
$$50 = FC$$
$$C = \frac{50}{F}$$

Therefore, they have the same indifference curve through the point $(10,5)$. See (b) for the graphs.

(b) The bundle gives Bridget and Erin utility levels of

$$U_B(15, 8) = 10(15)(8) = 1,200$$
$$U_E(15, 8) = 0.2(15^2)(8)^2 = 2,880$$

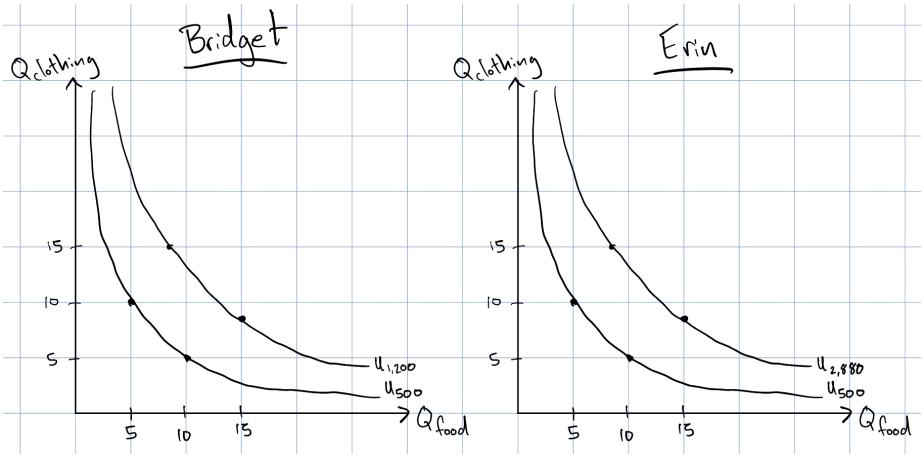
The bundles that give Bridget the same utility are defined by

$$1,200 = 10FC$$
$$120 = FC$$
$$C = \frac{120}{F}$$

The bundles that give Erin the same utility are defined by

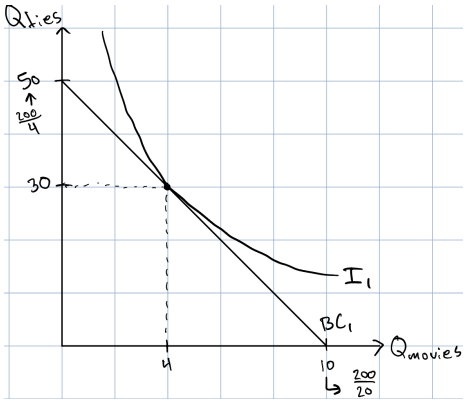
$$2,880 = 0.2F^2C^2$$
$$14,400 = F^2C^2$$
$$120 = FC$$
$$C = \frac{120}{F}$$

Once again, Bridget and Erin have the same indifference curve.

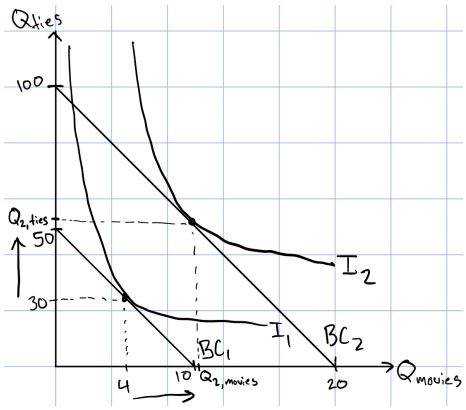



(c) I think they have the same preferences because they appear to have the same indifference curves. If you're looking for a challenge, it's possible to prove they have the same indifference curve running through an arbitrary bundle $(F_1, C_1)$.

4. (a) Julius's budget constraint is $200 = 20Q_M + 4Q_T$. At his optimal bundle, his MRS equals the (negative) slope of his budget constraint. Therefore, $MRS = 5$.

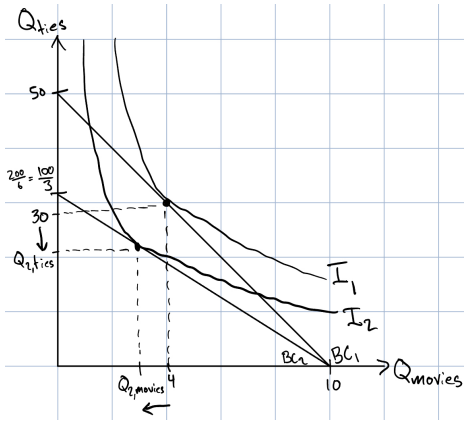



(b) I assume that movies and ties are normal goods, so when Julius's budget doubles, he's going to consume more of both.

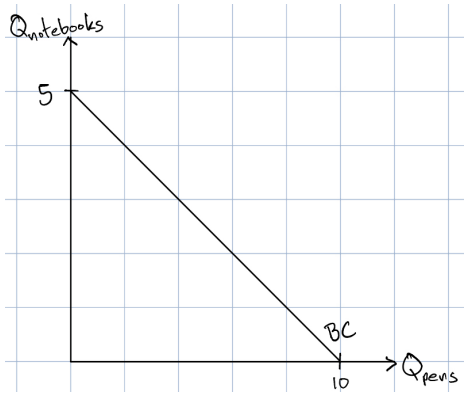



(c) The quantity of ties he consumes will unambiguously decrease. Since the relative price of ties increases, he will substitute away from ties and towards movies. Since his purchasing power decreases and ties are a normal good, he will consume fewer ties. The income and substitution effects are both negative.

As I've drawn it, the quantity of movies he consumes will decrease, but it's ambiguous. The income effect is negative because movies are a normal good, but the substitution effect is positive because movies become cheaper relative to ties. The overall effect of the price increase will depend on whether the income effect or the substitution effect is bigger. In my drawing, the income effect is bigger.




5. (a) Bernie can consume the bundles defined by $20 = 4Q_N + 2Q_P$.




(b)

| Quantity of Notebooks | Utility from Notebooks | Marginal Utility | Marginal Utility per Dollar |
|---|---|---|---|
| 0 | 0 | – | – |
| 1 | 32 | 32 | 8 |
| 2 | 60 | 28 | 7 |
| 3 | 84 | 24 | 6 |
| 4 | 104 | 20 | 5 |
| 5 | 120 | 26 | 4 |

| Quantity of Pens | Utility from Pens | Marginal Utility | Marginal Utility per Dollar |
|---|---|---|---|
| 0 | 0 | – | – |
| 2 | 28 | 14 | 7 |
| 4 | 52 | 12 | 6 |
| 6 | 72 | 10 | 5 |
| 8 | 88 | 8 | 4 |
| 10 | 100 | 6 | 3 |

(c) Bernie will buy 3 notebooks and 4 pens because that's the bundle that equalizes marginal utility per dollar across the two goods and that lies on his budget constraint.